

The Pace of AI Progress

What has changed, what the measurements mean, and what can reasonably follow from them.

Bryant Harrison • Murray State University

Evidence snapshot checked September 9, 2026. This is a selected historical collection, not a live leaderboard or complete account of AI research.

Learning goal

Participants should be able to describe a result with its task, metric, date, and conditions; separate a benchmark gain from a workplace outcome; and identify assumptions in a forecast.

Run a 20–25 minute session

1. **Measure • 10 minutes.** Start with “What has changed?” Explain that a benchmark is a defined test and scoring rule. Ask people to estimate a missing piece of history, then reveal it. Spend more time on the conditions than on whether an estimate was high or low.
2. **Context • 4 minutes.** Scrub the selected release timeline. Ask what the chart does and does not measure. Company lanes separate dots; height does not rank capability. Open the source-backed release list.
3. **Transfer • 5 minutes.** Compare the two workplace studies. Ask why an average gain for customer-support agents can coexist with slower completion in a specific developer study. Then inspect a scientific system.
4. **Forecast • 4 minutes.** Move the illustrative years slider. All three rules fit the same two invented points. Ask which additional evidence would help choose between them.
5. **Debrief • 2 minutes.** Have each participant name one task from their own work, one quality measure, and one failure they would watch for.

Controls and accessible participation

The Guide opens on each load; reopen it with **?**. Settings contains **Open Presenter Notes**, **Presentation mode**, and **Reset**. These notes are the same document as the printable guide. Reset clears the activity while preserving Presentation mode.

Draw with a finger or mouse. For a straight-line estimate, focus the chart and press Up or Down; Home and End set its endpoints. Choose any measurement from the selector. **Details** opens a side panel with an example, comparison limits, and a table. Hidden values appear in that table after Reveal. Escape closes a panel and returns focus.

Short route: use the coding, METR, and price measurements; compare the two work studies; end with the forecast slider. The other measurements remain available for discussion. No runtime network, AI inference, or storage is required.

Read each chart on its own terms.

1. Coding · SWE-bench Verified

The selected 2025 reports are Claude 3.7 Sonnet **63.7%**, Claude Opus 4 **72.5%**, GPT-5 **74.9%**, and Claude Sonnet 4.5 **77.2%**. The score is the share of selected repository issues resolved according to automated tests.

These are vendor-reported systems with different conditions. Claude 3.7 used 489 tasks; GPT-5 used a fixed 477-task subset; Sonnet 4.5 reports the full 500, averaged across ten trials with 200K thinking budget. Single-attempt wording does not establish like-for-like evaluation. Ask: “Which parts of the system changed besides the model?”

2. Scientific questions · GPQA Diamond

This is a **198-question** multiple-choice set. OpenAI reports GPT-4o **50.6%** and an o1 research model **77.3%** in its September 2024 comparison. Google's later table reports Gemini 3 Pro **91.9%** and Gemini 3.1 Pro **94.3%**, Thinking HIGH, no tools.

GPT-4o is located by its release date, with the later reporting date stated in Details. The o1 research point is an evaluation date, not the o1-preview product release. Vendors and compute protocols differ. Human markers from mismatched question sets are omitted. Ask: “Which parts of scientific work are not multiple-choice questions?”

3. Task duration · METR Time Horizon 1.1

The January 2026 report estimates **3.5, 60, 121, 101, 214, and 320 minutes** for the six selected models. These are human-expert task durations at predicted **50% AI success**, not AI runtime. The plot retains the reported 95% confidence intervals and the decline between o3 and Opus 4.

The tasks mostly concern software and research engineering. A later Sol estimate of 11.3 hours [5–40] is excluded from the curve: METR explicitly described the June 2026 estimates as non-robust. Details preserves that result and warning. A doubling-time fit depends on sample and date range; it does not prove acceleration or reliable autonomy.

4. Cost · A fixed MMLU threshold

Epoch's snapshot follows the cheapest listed API price meeting **MMLU $\geq 86\%$** . The rounded endpoints are **\$37.50** and **\$0.18**, about 200 times cheaper. Units are dollars per million tokens, with input and output prices blended **3:1**. The snapshot ends in February 2025.

This is not a price per million words or the full cost of a job. The threshold is one benchmark criterion, not “everything GPT-4 can do.” Ask what integration, review, retries, and token consumption would add.

A score belongs to a setup.

5. Puzzles · ARC-AGI-1 and a separate ARC-AGI-3 comparison

ARC Prize's December 2024 o3 preview evaluation reported **75.7%** at lower compute and **87.5%** at higher compute. The tuned preview was distinct from the later released o3. The points share a date because they are configurations, not sequential releases. Training on public practice examples and compute costs matter. The small gravity puzzle in Details is illustrative, not a representative test sample.

The next panel is deliberately separate. ARC-AGI-3 uses interactive games and **Relative Human Action Efficiency (RHAЕ)**. ARC Prize reported Astra at high reasoning effort with **54.8%** in a standard harness and **99.9%** in a provider harness, on the same semi-private set. Total reported costs were \$40,705 and \$18,817. This is not the percentage of ARC-AGI-1 puzzles solved, and it is not a continuation of that curve.

Prompt the room

"What would be missing if someone quoted only the model name and the larger score?" Look for the test version, metric, effort, harness, task set, and evaluation date.

Transfer: two work settings

Customer support. The revised Generative AI at Work study reports 5,172 agents and an average **15%** increase in issues resolved per hour. It used staggered deployment in a specific work setting. Benefits varied by worker; it is not a randomized estimate for all jobs.

Experienced developers. METR randomized AI access across **246 tasks** performed by **16 developers** on familiar open-source repositories. With early-2025 tools, AI-allowed tasks took **19% longer** (95% interval: 2–39% longer). Its February 2026 update suggests later improvement but says selection and timing limitations prevent a reliable magnitude estimate. Do not present the older number as today's effect.

Transfer: specialized scientific systems

GenCast. The 2019 evaluation beat ENS on CRPS in **97.2% of 1,320 comparisons** of variable, lead time, and level. That is not 97.2% weather forecast accuracy. Precipitation was excluded from the main results. Name the baseline and probabilistic scoring rule.

AlphaFold 3. Its structure-prediction results concern biomolecular complexes. Static structures do not simulate molecular dynamics or establish biological function. The paper describes limitations including stereochemistry and hallucinated structures. There is no single universal accuracy number in this activity.

Two points leave several futures open.

The scenario is invented. Its score rises from 20 to 30 in one year. Let t be years after the second observation. Three rules fit both anchors exactly:

Fixed gain: $30 + 10t$

Proportional gain: 30×1.5^t

Slowing toward a limit: $75 - 45 \times (9/11)^t$

At $t = -1$, all three give 20. At $t = 0$, all give 30. Farther out, their assumptions dominate. The slowing rule assumes a ceiling of 75; the score itself has a maximum of 100. Outputs above 100 are displayed as violations of the score's bounds, not silently clipped into plausible predictions.

Debrief questions

Which observation would help distinguish these rules? What might change the trend? What is the ceiling imposed by the test? What happens when the test becomes too easy? Which real task could you evaluate without assuming any of these futures?

Primary reading and verification boundaries

The in-app source list and data table carry full links. These starting points support the activity's measurement and transfer discussions; original authors and evaluators can still make errors.

- [Anthropic: Claude 3.7 Sonnet](#); [Claude 4](#); [Sonnet 4.5](#); [OpenAI: GPT-5](#). Read evaluation footnotes.
- [OpenAI: Learning to reason with LLMs, Appendix A](#); [Google: Gemini comparison table](#). Same benchmark name does not guarantee the same protocol.
- [METR: Time Horizon 1.1](#); [METR: Sol evaluation](#). Keep uncertainty and robustness warnings attached.
- [Epoch AI: Inference price trends](#). MMLU threshold and blended token pricing.
- [ARC Prize: o3 evaluation](#); [ARC Prize: Astra evaluation](#). Different benchmarks use different metrics.
- [Stanford: Generative AI at Work](#); [METR: Developer study](#); [2026 update](#). Populations and workflows matter.
- [Nature: GenCast](#); [Nature: AlphaFold 3](#). Specialized system results are not generic chatbot capabilities.

Checking the demo: Settings → Run data integrity checks tests source references, dates, bounds, and numerical behavior. It cannot validate a paper, prove causation, or establish that a selected catalog represents the field.